Building Unified 3D and 4D Content Generation (Application for Nectar Track in 3DV 2025)

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1 Motivation

The creation of high-quality 3D and dynamic 4D content represents a significant frontier in visual content generation, with potential applications in video games, visual effects, and wearable mixed-reality systems. While recent advancements in 2D content generation have transformed the field through innovations like diffusion models [8, 1, 3, 2] and autoregressive techniques [13, 10, 4, 7], extending these successes to 3D and 4D domains introduces new challenges. These include the inherent complexity of 3D modeling and the scarcity of datasets that capture both spatial and temporal dynamics, particularly in real-world scenarios.

My research focuses on developing a unified approach to 3D and 4D content generation. The core idea is to leverage both synthetic and real-world data to create a framework capable of modeling static 3D scenes and dynamic 4D content, where spatial and temporal information are seamlessly integrated. This direction aims to overcome limitations in current methods, which only rely on synthetic datasets [14, 5] and focus primarily on static 3D generation [6, 9, 11, 12].

2 Key Aspects

Data Curation for Dynamic Scenes: Exploring methods to curate a dataset of real-world videos with annotated camera poses and dynamic object information. This involves developing pipelines for detecting movable objects, estimating camera poses from static elements, and filtering for meaningful dynamics using motion-based criteria.

Spatio-Temporal Modeling Framework: Investigating a generative framework that disentangles spatial and temporal information while allowing flexible conditioning on single or multiple input views. This would involve designing modules capable of fusing multi-view and temporal data effectively, addressing the unique demands of 3D and 4D generation.

Democratizing Dynamic Content Creation: Prioritizing generalization to real-world scene-level generation, enabling users to enhance images and videos with richer, dynamic content.

3 Impact and Feasibility

This research direction has the potential to bridge critical gaps in 3D and 4D content generation, advancing both foundational understanding and practical applications. By addressing challenges such as dataset scarcity and spatio-temporal integration, this work aims to lay the groundwork for new tools and methods that democratize dynamic content creation. Anticipated contributions include scalable methods for data collection, innovative generative modeling techniques, and a publicly available dataset to foster broader research and collaboration.

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